Yongjun Rong
Cherry Hill, NJ, USA, 08003 (US Citizen) **Email:** rongyj@hotmail.com **Tel:** 856-408-4027
Linkedin: https://www.linkedin.com/in/rongyongjun/, **Github:** https://github.com/rongyj

EDUCATION

Jun.,2006, Master of Computer Science, Texas Tech University, Lubbock, TX, USA

Feb.,1996, Master of Engineering, Shanghai JiaoTong University, Shanghai, China

Jul.,1991, Bachelor of Engineering, Changsha Railway University, Changsha, Hunan, China

SUMMARY:

Yongjun Rong is a visionary cloud architect and technology leader with deep expertise in GenAI systems, prompt engineering, cloud infrastructure, and digital transformation. With over two decades of experience spanning finance, consulting, SaaS, and technology sectors, he has architected and delivered large-scale distributed systems, led mainframe-to-cloud migrations, and pioneered automation and DevOps best practices. Yongjun excels at designing resilient, scalable microservices, integrating enterprise SaaS applications, and driving innovation through mentorship and cross-team enablement. His technical breadth covers AWS, GCP, Azure, Java, Python, Node.js, Kubernetes, Terraform, and more. Fluent in both English and Chinese, he is passionate about empowering organizations to achieve agility, reliability, and business impact through modern cloud-native solutions.

SKILLS:

- GenAI system architecture and implementation with Retrieval-Augmented Generation (RAG) and multi-agent orchestration to accelerate software development and mainframe migration to modern cloud-native platforms.
- Cloud infrastructure design, automation, and migration across AWS, GCP, and Azure, including serverless architectures (Lambda, EventBridge, Step Functions), container orchestration (EKS, AKS, GKS), and Infrastructure as Code (Terraform, Terragrunt, CloudFormation, Ansible, Puppet, Chef).
- Enterprise architecture leadership for large-scale finance, insurance, and telecommunications organizations, specializing in distributed systems, microservices, event-driven architectures, streaming data processing, and technology strategy development.
- Digital transformation engineering, refactoring legacy monolithic systems to cloud-native, loosely-coupled distributed architectures using Java, Python, Node.js, SpringBoot, and modern data stores (SQL, NoSQL, DynamoDB, MongoDB, ClickHouse, PostgreSQL, Oracle).
- DevOps and automation expertise in cloud and on-premises environments, including CI/CD pipeline design, configuration management, system/ network administration, and orchestration with Docker, Kubernetes, Jenkins, Bamboo, and Linux HA technologies.
- Integration architecture and implementation for SaaS enterprise applications (ERP, CRM, Accounting, HR), leveraging VPC Endpoints, PrivateLink, REST/SOAP APIs, WSDL/XML/JSON, and secure authentication/authorization mechanisms.
- Design and delivery of highly available, scalable, and resilient microservices and event-driven systems, applying best practices in CQRS, event sourcing, domain-driven design, and cloud-native patterns for internet-scale applications.
- Mentorship and leadership in technology adoption, process automation, and cross-team enablement, fostering a culture of innovation, reliability, and continuous improvement across engineering organizations.

HOBBIES

- Pickleball (~4.0 player)
- Pingpong (ITTF Score around 1200)
- IoT or Robot programming (Have iRobot and MIPOSaur at home. Programming against iRobot create SDK and MIPOSaur SDK.)
- iOS/Android Apps Development (Developing a Chinese character learning App)
- Playing with cutting-edge technologies (genAI/Autogen/Multiagents, Docker, Go, Kafka, Kubernetes and Amazon CloudFormation, JavaScript, AngularJS, ReactiveJS, Node.js)
- IFTTT Home Automation (IFTTT, WeMo, EcoBee, Weather Underground, SaaS, REST, Home Automation)

WORK EXPERIENCE SUMMARY:

Employments	Major Accomplishments & Technologies
Feb.,2025 – Current **Biophy** Cherry Hill, NJ (WFH),USA **Director of Cloud Architecture**	• Comprehensive Infrastructure as Code (IaC) Design and Implementation • AI-Powered Data Ingestion and Scraping System for RAG • Custom AWS CloudWatch Alerting System with EventBridge, Lambda, SQS and SNS
	IaC, Terraform/Terragrunt, AWS/EC2/EBS/S3/VPC/ALB/Security/R53/IAM/Secrets Manager/Lambda/CloudWatch/EventBridge/SQS/SNS, Python/DJango/Pydantic, openAI API/LangChain, SOLID Principles, Clean Architecture,
Dec.,2021 – Feb.,2025 **McKinsey & Company** Cherry Hill, NJ (WFH),USA **Senior Cloud Engineer II (Associate)**	• Using genAI (openAI) with RAG (ChromaDB, Milvus) to accelerate the development of data pipelines running on pyspark and databricks • Using genAI (openAI) with multi-agents to accelerate the migration of mainfarme to modern technologies • Migration/Refactoring of Mainframe (IBM MQ) EDA system to AWS by using EventBridge/Lambda/Step Function/CloudWatch/EKS/SQS/SNS...etc.
	genAI/openAI/Langchain/RAG/Multi-agents/prompts tuning, VectorStore/ChromaDB/Milvus, Python/Flask/FastAPI/Pydantic/Jupyter, Databricks/pyspark, Typescript/JavaScript/NodeJs, Event Driven Architecture (EDA), Terraform, Microservices, Java/SpringBoot/NodeJs/Python/Flask/FastAPI, EventBridge/CloudWatch/Lambda/Step Functions, DynamoDB/RDS/Oracle, Docker/K8s/EKS/AKS, Json Schema, JOI, OpenAPI/Swagger,
Dec.,2020 – Dec., 2021 **Yotascale** Cherry Hill, NJ (WFH),USA **Staff Engineer**	• Lead Architect and Lead Implementation Engineer for Cloud(AWS/Azure/GCP) Cost Analytics/Optimization/Management platform
	OpenAPI/Swagger, Microservice, Java/Kotlin/SpringBoot, Kubernetes/EKS/AKS, Istio, Terraform, CQRS, Event Store, Event Driven Architecture (EDA), Kafka/ActiveMQ/IBM MQ, Kafka Stream, Docker, AWS(EC2/IAM/S3/ECS/EKS/ELB/Lambda/RDS/VPC/R53/Secret Manager/AutoScaling/SES..etc), Puppet/Chef/Ansible, Git/Maven/Jenkins, Python/Bash/Go, ELK/AppDynamic/Datadog,
Mar.,2020 – Nov., 2021 **Gigster Inc.** Cherry Hill, NJ,USA **Technical Infrastructure/ Devops Architect**	• Images manipulation/detection software evaluation system
	Microservice, NodeJs, Kubernetes/EKS, Terraform, RabbitMQ/ActiveMQ, Docker, AWS(EC2/IAM/S3/EKS/ALB/RDS/VPC/R53/AutoScaling/SES/ECR..etc), Puppet/Chef/Ansible, Git/Maven/Jenkins, JavaScript,
Jun.,2017 – Feb., 2020 **Barclays** Wilmington, DE,USA **VP,Lead Architect Engineer**	• Delivery Lead for strategic distributed system technology domain • Group-wide Cloud Adoption for Contact Center, Merchant Services and Fraud
	Terraform, CQRS, Event Store, Event Driven Architecture (EDA), Kafka/ActiveMQ/IBM MQ, Kafka Stream, Docker, AWS(EC2/IAM/S3/ECS/EKS/ELB/Lambda/RDS/VPC/R53/Secret Manager/AutoScaling/SES..etc), Puppet/Chef/Ansible, Git/Maven/Jenkins/Bamboo, Python/Ruby/Scala/Bash/Go, ELK/AppDynamic, DynamoDB/MongoDB/ElasticCache/GridGain, Linux/Windows/AIX/Solaris, aPaaS/VMWare/Citrix,
May, 2014 – June, 2017 **Oracle** Redwood, CA (WFH@NJ),USA **Lead Software Engineer**	• Docker Swarm/Compose/Stack CD/CI Test Environment for Cloud SDK/APIs/Connectors • Cloud Enterprise Applications Integration(EAI) connectivity SDK/API plugin framework for Oracle SOA Suite • Oracle Fusion Applications (FA) Business Event/Message-Driven Connector Framework
	Java, SOA, Web Services, SOAP/WSDL, REST/JSON/XML, XSchema/XSD, DOM, WSDL4J, SDK, API, MVC, Docker, Cloud, Amazon EC2, NoSQL, GIT, Maven, Junit, SoapUI, IaaS, SaaS, AWS,
Oct. 2010 – Apr. 2014 **Comcast T&P** Philadelphia, PA,USA **Senior Cloud DevOps/ Infrastructure Automation Engineer**	• AutoBuild System for Build/Release more than 200 Microservice projects with GIT/SVN/Maven • Configuration management and deployment automation with Puppet and Capistrano • Continuous Integration (CI) and Continuous Deployment (CD) with Bamboo and Jenkins • Performance Monitoring System for Jetty, JVM, Linux and Splunk • Secured FTP server with virtual users and admins users with complex access permissions • BOM (Bill Of Materials) Administration and Management Application (BAMA) Enhancement and maintenance • Two Steps bare-metal Linux bootstrap system for HP Servers • Linux HA for Git Repo(Gerrit) and Maven Repo (Artifactory/Nexus) Servers • Automation scripts and tools for DevOps • CoreOS and Docker prototyping and testing • SDLC process management and junior team member training and mentoring
	DevOps, Automation, Docker, CoreOS, Ruby, Ruby on Rails 3, Bash, SDLC, GIT, Maven, Jira/Confluence, Bamboo/Jekins/Hudson, Nexus/Artifactory, Gerrit, CI/CD, Performance Monitoring, Splunk, JMX, Capistrano, PXE/TFTP, DHCP, DNS, Cobbler, Centos, MySQL/PostgreSQL, NoSQL, Cassandra, RabbitMQ, JMS, LDAP, SSO, Metrics, Graphite, Agile, RPC, Cloud, Clustering, Linux HA, HAProxy, Pacemaker/Corosync/Cman, SAN, GFS2, RAC, Volume Manager,
Nov., 2006 – Oct., 2010 **Boomi Inc.** Berwyn, PA,USA **Senior Software Engineer**	• Boomi On-Demand Business Integration Cloud Service • Cloud Platform/Infrastructure Design and Implementation for Boomi On Demand Production Environment • Desktop Accounting and ERP Enterprise Applications Integration (EAI) • Amazon EC2 and S3 SOA HA Environment • Setup Development, Build and Deployment Environment for QA, Staging and Production Environment • Research virtualization technology and implement Virtual Machines using VMware server and workstation • Java to C/C++, DLL, COM/COM+/DCOM/.NET Integration • SaaS system integration • SAP system integration

	Java, SOA, IaaS, PaaS, EAI, XML, XSD, JAXB, JavaScript, .NET, EC2, S3, EBS, REST, SOAP, WSDL, SaaS, SAP, BAPI, ABAP, MySQL/PostgreSQL, Maven, SVN, JVM, JVM GC Tuning, Apache Lucene, Apache Solr, VMWare, Cloud,
Jun., 2006 – Nov.,2006 **Vantage learning** Newtown, PA,USA **Senior System Engineer Manager**	• Large scale SOA e-learning system architecture design and blueprinting • Large Data Center Design and Administration • Sun SPARCStorage SSA Array 100 and Sybase SQL Server (10.x, 11.x) Data (DB schema, data and Stored-Procedure) Recovery • Ecommerce Open Source Software installation and Configuration • Resin Application Server Distributed Session Design and Setup
	Linux, Centos, SOA, Java, .NET, Network, VLAN, Router, Firewall, Load Balancer, SOAP/WSDL, MySQL/PostgreSQL, Struts, Disk Array, Fedora, Redhat Linux, LAMP, eCommerce, SQL,
May, 2002 – Jun.,22,2006 **Computer Science, Texas Tech University** Lubbock, Texas,USA **Senior Programmer/ Analyst and Unix System Administrator**	• Unix network System and Software routine maintenance • Setup Veritas Netbackup System with RAID0/1/5 and Tape Library • Setup LDAP, Kerberos KDC and Cross-Realm system • Migrated email system from UNIX mail server to Windows Exchange Server • Secured Postfix SMTP server with SASL Authentication and Verification • Installed AIX 5.1 and configured NIM on IBM RS6000 (10x 7043-140 boxes and 1x 7024-E30 box) • Deployed Globus Tookit3.2.1 (OGSA/OGSI) Grid Computing Environment on Solaris 9(3 boxes) and Linux (3 boxes) • Setup Jumpstart Environment for Solaris 9 and Solaris 10 • Designed and Implemented the Single Sign-On (SSO) for Unix and Windows Environments • The Migration from NIS+ to LDAP for Solaris
	JAVA, C/C++, GCC, GNU Make, perl, LDAP, NFS, SSO, Kerberos, SSL, SMTP, Email Server and Client, Solaris, AIX, Mac OSX, Linux, Windows Server, PAM, RAID, SCSI, Veritas, TFTP, RARP/ARP, Bootparamd, DHCP, Postfix, PostgreSQL, NIS/NIS+,
Aug.,2002 – Jun.,2006 **Computer Science, Texas Tech University** Lubbock, Texas,USA **Master Student**	• Master Thesis: A Service Wrapping and Provisioning Framework for SOA • GIS Project: A Yield Mapping and Prediction Information Delivery System • Independently finished practical course projects list
	Java, C/C++, GCC, Gnu Make, Linux, Linux Kernel, GIS, Ant, Struts, J2EE, UML2, MVC, Maven, PostGIS, PostgreSQL, Solaris,

SELECTED PROJECTS

1. Feb.,2025 – Current, Director of Cloud Architecture, Biophy , Cherry Hill, NJ (WFH),USA

As a hands-on director of cloud architecture, I designed and implemented comprehensive infrastructure as code (IaC) by importing existing AWS resources and redesigning them according to AWS best practices. I led the development of a data ingestion and scraping system for an AI-powered RAG platform, following SOLID principles and Clean Architecture patterns. Additionally, I created a custom AWS CloudWatch custom alerting system leveraging Lambda, SQS and SNS for robust monitoring and notification.

1.1 Comprehensive Infrastructure as Code (IaC) Design and Implementation

Led the transformation of AWS infrastructure to infrastructure as code (IaC), importing and redesigning resources to align with best practices. Developed modular Terraform configurations and established secure, standardized cloud operations.

Highlights:

- Imported and catalogued existing AWS resources including EC2 instances, VPC configurations, security groups, and IAM policies
- Redesigned infrastructure architecture following AWS Well-Architected Framework principles
- Implemented modular Terraform configurations with reusable modules for consistent deployment
- Established comprehensive security controls including least-privilege IAM policies and network segmentation
- Created automated validation and testing procedures for infrastructure changes
- Documented infrastructure patterns and best practices for team adoption

Technologies:

Terraform, AWS EC2/EBS/S3/VPC/ALB/Security/R53/IAM/Secrets Manager, Infrastructure as Code, AWS Best Practices, Security Controls,

Leadership:

- Led the transformation from manual infrastructure management to fully automated IaC practices.
- Established infrastructure standards and patterns that improved team productivity and security posture.
- Mentored team members on AWS best practices and Terraform/Terragrunt development.

1.2 AI-Powered Data Ingestion and Scraping System for RAG

Architected and delivered a robust data ingestion and scraping platform for AI-powered RAG, applying SOLID and Clean Architecture principles to ensure scalability, maintainability, and high data quality.

Highlights:

- Architected a modular data ingestion pipeline supporting multiple data sources and formats
- Implemented intelligent web scraping capabilities with rate limiting and error handling
- Designed data transformation layers following Clean Architecture principles
- Built robust error handling and retry mechanisms for data processing failures
- Implemented comprehensive logging and monitoring for data pipeline health
- Created automated testing frameworks for data quality and pipeline reliability

Technologies:

Python/Django/Pydantic, openAI API/LangChain, SOLID Principles, Clean Architecture, Data Processing, Web Scraping,

Leadership:

- Drove the adoption of Clean Architecture and SOLID principles across the data engineering team.
- Established data quality standards and monitoring practices for AI system reliability.
- Mentored developers on modern software design patterns and best practices.

1.3 Custom AWS CloudWatch Alerting System with EventBridge, Lambda, SQS and SNS

Developed a proactive monitoring and alerting system using AWS CloudWatch, Lambda, SQS, and SNS, enabling automated detection and response to critical events across cloud infrastructure.

Highlights:

- Designed custom CloudWatch metrics and dashboards for comprehensive system visibility
- Implemented SQS-based message queuing for reliable alert processing and delivery
- Configured SNS topics and subscriptions for multi-channel notifications (email, SMS, Slack)
- Built automated response workflows triggered by specific alert conditions
- Created alert escalation policies based on severity and time-based thresholds
- Implemented alert correlation and deduplication to reduce alert fatigue

Technologies:

AWS CloudWatch, SQS/SNS, Lambda, EventBridge, Monitoring, Alerting,

Leadership:

- Established comprehensive monitoring and alerting standards across the organization.
- Reduced mean time to detection (MTTD) and mean time to resolution (MTTR) through proactive monitoring.
- Trained operations teams on monitoring best practices and incident response procedures.

2. Dec.,2021 – Feb.,2025, Senior Cloud Engineer II (Associate), McKinsey & Company , Cherry Hill, NJ (WFH),USA

Led the development of genAI systems with RAG and multi-agents to accelerate mainframe migration to modern technologies. Built comprehensive genAI solutions with RAG to speed up data pipeline development on PySpark and Databricks for complex data transformations. Scaled RAG vectorstore (ChromaDB) for a call center chatbot copilot. Successfully migrated and refactored complex EDA mainframe systems to AWS serverless architecture using EventBridge, Lambda, Step Functions, SQS, SNS, CloudWatch, and EKS.

2.1 Using genAI (openAI) with RAG (ChromaDB, Milvus) to accelerate the development of data pipelines running on pyspark and databricks

Designed and implemented a genAI system with RAG (ChromaDB/Milvus) to automate and accelerate the development of data pipelines, enabling seamless transformation of source tables to target tables with varying data structures.

Highlights:

- Prompts tuning to generate data pipelien pyspark function code and unitest code
- Design and implement chunking system (by target table or by relationship) to split the large inputs into small chunks to avoid overflowing the openAI API token limit
- Design and implement parallel calls to openAI API by using asyncio with yield to improve performance
- Implement FastAPI API to expose source to target data transformation as REST service
- Design and document the local development and debug workflow by using debugpy with VSC and pycharm

Technologies:

Langchain/VectoreStore/ChromaDB/Milvus, genAI/openAI, RAG, Python/Pydantic/Asyncio/Flask/FastAPI/Debugpy, Typescript/JavaScript/NodeJs, Microservice, Dockerfile/Docker/Compose, PostgreSQL,

Leadership:

- Spearheaded the integration of genAI and RAG for data pipeline automation, enabling rapid transformation of complex data structures.
- Mentored engineers on advanced prompt engineering and parallel AI workflow design, raising team proficiency in cutting-edge AI techniques.
- Established robust local development and debugging standards, improving team productivity and onboarding efficiency.

2.2 Using genAI (openAI) with multi-agents to accelerate the migration of mainfarme to modern technologies

Built a multi-agent genAI platform to automate and accelerate the migration of mainframe applications to modern technologies, integrating RAG for context-aware transformation and process automation.

Highlights:

- Core Contributors to genAI system to migrate mainframe frontend technologies (BMS/JSF/JSP) to modern technologies (React/Angular..etc.)
- Contribute to develop multi-agents genAI system to automate the development/testing/deployment process
- Contribute to develop RAG system to accelerate the migration process from Cobol to Java by using ChromaDB/Milvus to store the dendepncy context

Technologies:

Langchain/VectoreStore/ChromaDB/Milvus, genAI/openAI, RAG, Agents/Multi-agents/Autogen, Python/FastAPI, Typescript/JavaScript/NodeJs, Microservice, Dockerfile/Docker Compose, PostgreSQL,

Leadership:

- Drove the integration of multi-agent genAI systems to automate legacy migration, reducing manual effort and error rates.
- Collaborated across frontend and backend teams to ensure seamless modernization from mainframe to cloud-native stacks.
- Championed the use of RAG for context-aware migration, setting a new benchmark for future modernization projects.

2.3 Migration/Refactoring of Mainframe (IBM MQ) EDA system to AWS by using EventBridge/Lambda/Step Function/ CloudWatch/EKS/SQS/SNS...etc.

Engineered and deployed an EKS-based router to integrate AWS EventBridge with external MQ events, providing robust monitoring, reporting, and cloud-native migration for mainframe EDA systems.

Highlights:

- Develop SpringBoot EKS router to convert XML/XSD to JSON schema and integrate with MQ events
- Design and implement the monitoring terraform module for all different AWS resources (Lambda, EventBridge, S3, DynamoDB, Step Function..etc.) using cloudwatch metrics
- Design and implement the reusable report generator using Lambda and EventBridge and deployed as Microservices which can be called to run any custom AWS CloudWatch insights queries in custom schedule

Technologies:

Java/SpringBoot, Terraform, Microservice, Java/JavaScript, EKS, AWS/EventBridge/CloudWatch/Lambda, XML/XSD/Json/Json Schema, DynamoDB, RDS/Oracle,

Leadership:

- Led the end-to-end migration of mission-critical EDA systems to AWS, ensuring zero downtime and robust cloud integration.
- Developed reusable infrastructure modules, empowering other teams to accelerate their own cloud migration journeys.
- Instituted comprehensive monitoring and reporting standards, elevating operational visibility and reliability.

3. Dec.,2020 – Dec., 2021, Staff Engineer, Yotascale , Cherry Hill, NJ (WFH),USA

Led the architecture and development of a comprehensive cloud cost analytics, optimization, and management platform supporting AWS, Azure, and GCP. Designed and implemented cross-cloud cost visibility, allocation, and optimization capabilities to help organizations manage cloud spending effectively.

3.1 Lead Architect and Lead Implementation Engineer for Cloud(AWS/Azure/GCP) Cost Analytics/Optimization/ Management platform

Architected and implemented a cross-cloud cost analytics and optimization platform, enabling organizations to gain actionable insights and manage cloud spending across AWS, Azure, and GCP. Led the design of scalable data pipelines, analytics microservices, and secure infrastructure.

Highlights:

- Introduce and Build Microservice Design First Methodology and team culture based on OpenAPI/Swagger Specification
- Design and develop Cost Analytics/Lens Navigation/Lens Management Microservice using SpringBoot with ClickHouse and PostgreSQL as DW and DBMS
- Design and implement dynamic metadata driven widget/filter configuration Microservice
- Implement bitbucket CI/CD pipeline to build, deploy Microservices/GraphQL to eks/k8s with Istio
- Design the AWS infrastructure architecture with EKS/ALB/Istio/S3/SES/R53/ECR/PostgreSQL/RabbitMQ/SFTP
- Design/Implement terraform templates/scripts to deploy the full AWS infrastructure stack with IAM roles/users
- Develop terraform templates to deploy ALB controller/RabbitMQ and PostgreSQL by using helm provider
- Develop terraform templates to deploy k8s' CRDs, Pods, Services, Service Account by using Kubernetes-alpha and Kubernetes providers

- Design/Develop multiple terraform modules with Terragrunt.hcl to maintain the inputs/outputs dependencies between modules and keep configurations DRY
- Develop terraform templates to dynamically add R53 zone and records with TLS certificates validations
- Develop terraform templates to dynamically create S3 buckets with IAM policies to allow k8's service account to assume IAM role to access control using OIDC
- Develop terraform templates to create SFTP server backed with S3 bucket with IAM role and policy for sftp users
- Compile Microservice design/development guide lines based on OpenAPI/Swagger Spec
- Implement Junit test cases with local docker test containers
- Develop common multiple DBMS (ClickHouse and PostgreSQL) access lib with AWS secrets manager
- Implement json schema validation using open source

Technologies:

Cloud Cost Analytics/Optimization/Management, OpenAPI/Swagger, Microservice, Java/Kotlin/Python, Spring/SpringBoot/Flask, Kubernetes/EKS/AKS/GKS, Terraform/Terragrunt/Ansible, AWS/EC2/EBS/S3/VPC/Security/R53/IAM/Secrets Manager/Lambda, Service Mesh/Istio/Envy, Helm/Chart, RabbitMQ, Azure/GCP, Junit/Test Containers, Docker/Minikube,

Leadership:

- Architected a cross-cloud cost analytics platform, aligning technical vision with business goals for multiple stakeholders.
- Introduced and enforced microservice design-first culture, elevating team standards and delivery quality.
- Mentored engineers in infrastructure-as-code and CI/CD best practices, fostering a culture of automation and reliability.

4. Mar.,2020 – Nov., 2021, Technical Infrastructure/Devops Architect, Gigster Inc. , Cherry Hill, NJ,USA

Designed and implemented comprehensive AWS infrastructure architecture including CI/CD automation, Terraform infrastructure as code, and EKS container orchestration. Led the development of scalable, automated cloud infrastructure solutions to support microservices and distributed systems.

4.1 Images manipulation/detection software evaluation system

Led the architecture and implementation of an image manipulation and detection evaluation platform using microservices and event-driven architecture on AWS EKS. Automated infrastructure provisioning and deployment pipelines for scalable, efficient operations.

Highlights:

- Design and implement AWS infrastructure for Microservices (NodeJs) and EDA within EKS cluster
- Design and implement fully automated AWS stack in IaaS git repo by using terraform/terragrunt/bash
- Develop terraform scripts/modules to dynamically deploy/configure multiple Microservices with pods/containers/services to EKS cluster by using kubernetes terraform provider
- Develop terraform scripts to deploy/configure AWS ALB as eks ingress controller by using helm terraform provider
- Develop terraform scripts to customize IAM role/policy, k8s SA(service account) and CRD (Custom Resources Descriptor) for ALB Ingress Controller by using Kubernetes-alpha terraform provider
- Develop terraform templates to deploy RabbitMQ and PostgreSQL by using terraform helm provider
- Design/Develop multiple terraform modules with Terragrunt.hcl to maintain the inputs/outputs dependencies between modules and keep configurations DRY
- Develop terraform templates to dynamically add R53 zone and records with TLS certificates validations
- Develop terraform templates to dynamically create S3 buckets with IAM policies to allow k8's service account to assume IAM role to access control using OIDC
- Develop terraform templates to create SFTP server backed with S3 bucket with IAM role and policy for sftp users
- Develop terraform templates to dynamically create IAM user with customized group and policies

Technologies:

Cloud Cost Analytics/Optimization/Management, OpenAPI/Swagger, Microservice, Java/Kotlin/Python, Spring/SpringBoot/Flask, Kubernetes/EKS/AKS/GKS, Terraform/Terragrunt/Ansible, AWS/EC2/EBS/S3/VPC/Security/R53/IAM/Secrets Manager/Lambda, Service Mesh/Istio/Envy, Helm/Chart, RabbitMQ,

Leadership:

- Designed and implemented a fully automated cloud infrastructure, setting a new standard for deployment efficiency.
- Enabled rapid scaling and reproducibility by modularizing infrastructure as code, empowering future project teams.
- Provided hands-on mentorship in DevOps automation, upskilling team members in Terraform and Kubernetes.

5. Jun.,2017 – Feb., 2020, VP,Lead Architect Engineer, Barclays , Wilmington, DE,USA

Directed the cloud adoption journey, migrating on-premises enterprise systems to AWS. Served as delivery lead for distributed system technology, including microservices, event-driven architecture, message transport, integration (batch/ETL), and event stream processing.

5.1 Delivery Lead for strategic distributed system technology domain

Directed the architecture and technology strategy for distributed systems, microservices, event-driven architecture, and stream processing, driving innovation and technology transformation across the organization.

Highlights:

- End to End Architecture; guiding solution architecture; developing design patterns Architecture Reviews - identifying reuse opportunities, manage technical debt Monitoring emerging technologies, industry trends and their applicability to the enterprise Capability roadmap and heatmap Drive/Foster Innovation Influencing Strategic Planning
- Drive technology transformation from on-premises to public cloud, multi-tiers monolithic to cloud-native Microservice system, strong ACID to BASE data consistency, Sync/Request/RPC driven to Async/Event driven architecture, tight coupling to loose coupling, ETL/Batch/Time to realtime streaming processing, vendor locked to resilient commodity system, vendor favor to OSS, vertical scaling to horizontal scaling
- Establish group-wide distributed system architecture and technology strategy
- Build PoC and Reference Architecture for common use cases for distributed system

- Evaluate around 200 existing products/technologies and decide the invest/divest house positions
- Create the architecture handbook for Microservice as group-wide guidance for solution Architect
- Contribute to and review the architecture handbook for EDA, Stream Processing and Integration
- Design vendor and technology selection and design principles/standards and best practices for the domain
- Review and approve the new request for product/technology in the domain
- Maintain the scope document and product list for the domain
- Develop the roadmap to adopt new products/technology and decommission legacy products/technology
- Lead working group (~30 members) forum to discuss and make decision on technology strategies

Technologies:
Technology strategic Design and Stewardship, Distributed system, Domain Driven Design, Microservice, ACID/BASE, (Event) Stream /Batch Processing, Event Driven Architecture (EDA), Message Transports (Kafka, ActiveMQ, IBM MQ), Data Integration (ETL/ELT), Compensating transactions/Sagas, JMS/AMQP, Resilience and Performance, CQRS/SOA, CAP,

Leadership:
- Directed a cross-functional working group to define and execute the distributed systems technology strategy.
- Authored the architecture handbook for microservices, influencing solution design across the organization.
- Drove technology transformation initiatives, guiding teams through cloud adoption and event-driven architecture.

5.2 Group-wide Cloud Adoption for Contact Center, Merchant Services and Fraud

Led cloud assessment and migration strategy for hundreds of on-premises applications, providing technical guidance and troubleshooting to ensure successful adoption of AWS cloud architecture.

Highlights:
- Build python automation tools to access AWS secret manager via boto3
- Using python boto3 to build tools to dynamically manipulate the IAM policies, roles and users
- Using terraform to dynamically deploy whole AWS infrastructure including EKS, IAM, DNS, ALB, S3 ...etc.
- Building and maintaining the EKS cluster via terraform and k8s yaml
- Defined and designed cloud strategy and roadmap/journey for Merchant services, Fraud and Contact Center
- Ensured ~500 on-premises application cloud migration journey
- Assess ~500 on-premises applications to determine the cloud migration readiness
- Provide guidance to application team to design cloud migration architecture
- Guide the .NET system to AWS migration
- Consulting to migration on-premises system to AWS ECS
- Collect system data from Abacus, ServiceFirst query and repo tools

Technologies:
AWS, EC2, RDS, S3, ECS, VPC, ACL/Security Group/Routing table, ELB/NLB/ALB, Private Link, EBS,

Leadership:
- Led the assessment and migration of 500+ on-premises applications to AWS, ensuring business continuity.
- Developed cloud migration roadmaps and provided hands-on guidance to application teams.
- Championed automation of cloud infrastructure, reducing manual errors and accelerating delivery.

6. May, 2014 – June, 2017, Lead Software Engineer, Oracle , Redwood, CA (WFH@NJ),USA

The Cloud Connectivity software engineering team is responsible to develop the Cloud Integration Connectivity SDK and Java Connectors which enable the SOA Suite and Oracle Integration Cloud Service (ICS) connectivity to any Oracle internal or external SaaS or On-Premises Enterprise Applications.

6.1 Docker Swarm/Compose/Stack CD/CI Test Environment for Cloud SDK/APIs/Connectors

Pioneered the use of Docker Swarm, Compose, and Stack to create scalable, automated CI/CD test environments for cloud SDKs, APIs, and Java connectors, streamlining development and testing workflows.

Highlights:
- Designed and setup complex Docker containers network with Overlay and MACVLAN drivers
- Created Dockerfile to build WebLogic docker images with different external modules and libraries
- Built Docker Hosts in Oracle Cloud VMs, Amazon EC2 and local VirtualBox by using Docker Machine
- Designed and built the pool of Docker Cluster Environment by using Docker Swarm
- Created Docker Stack file and deploy the service stack for multiple Microservice
- Developed docker-compose.yml files with extending services for the full service stack including dependency Web Services, DBMS, NoSQL and JMS
- Researched and prototyped the Docker Swarm and Docker Compose integration

Technologies:
Docker, Docker Machine, Docker Compose, Docker Swarm, Docker Stack, Docker CLI (Rest APIs), WebLogic, Oracle Cloud VMs, AWS, JMS, NoSQL (Cassandra), DBMS (MySQL and PostgreSQL),

Leadership:
- Pioneered the adoption of containerization for CI/CD environments, reducing setup time and increasing test reliability.
- Developed reusable Docker orchestration templates, enabling rapid prototyping for SDK and connector teams.
- Trained team members on advanced Docker networking and orchestration concepts.

6.2 Cloud Enterprise Applications Integration(EAI) connectivity SDK/API plugin framework for Oracle SOA Suite

Developed a robust SDK and plugin framework for seamless integration between Oracle SOA Suite and a wide range of SaaS and on-premises enterprise applications, enhancing connectivity and extensibility.

Highlights:

- Designed and implemented the WSDL regeneration framework by rebuilding the internal data model (DOM) from the persisted integration flow artifacts files (JCA, WSDL and XSD) and the remote custom data
- Designed and implemented new APIs for the SDK to provide cache and pagination abilities for the backend persistent layer
- Designed and implemented NLS and localization framework for displayable strings within the SDK
- Troubleshoot the WSDL/XSD generation issues related to complex mixing namespaces and prefixes
- Develop utility methods to handle the namespaces and prefixes conflicts while combining the importing external schema to a single WSDL file
- Develop recursive logic to follow the XML element types to embed the undefined data structure if it has the same target namespace or create new schema if it has different target namespace
- Dynamically and Programmatically generate or regenerate the simplified integration WSDL from large set of source WSDL, schema XSD files and fresh custom data by using XDK/JAXP/WSDL4J/DOM
- Programmatically manipulate the single WSDL file generation from the source WSDL and schema XSD files with external schema file importing
- Programmatically handle the anonymous Complex and Simple XSD type for the integration WSDL generation and regeneration
- Design and test Oracle RCU DB descriptive XML files for new DB tables/sequences with primary and foreign keys
- Worked together with different connector development teams to joint debug the WSDL regeneration related issues
- Train and transfer knowledge for the WSDL regeneration framework to geographically distributed connector development teams

Technologies:

Java, SOA, SOAP/REST, WSDL/JSON, XML, XML Schema, API, SDK, IaaS, RCU, JCA, XDK, JAXP, WSDL4J, DOM, WS-Security, Jersey REST/JSON, MVC, ADF, NLS/L10, Oracle DB, Jira, Agile, ADE, GIT, Maven, Junit, XMLUnit, SoapUI, OSC, ERP, HCM, RightNow,

6.3 Oracle Fusion Applications (FA) Business Event/Message-Driven Connector Framework

Engineered a business event and message-driven connector framework for Oracle Fusion Applications, enabling real-time event discovery, filtering, and integration with external systems.

Highlights:

- Design and implement the business event connector for FA (OSC/ERP/HCM)
- Build the design time web UI using the Oracle Web UI SDK which is building on the top of ADF
- Design and Implement the Business Event Connector Runtime
- Programmatically manipulate the payload to align with the changed schema in the integration WSDL
- Design and implement the custom event based on the generic event concepts
- Programmatically replace the xsi:anyType generic object type with the concrete object type selected by the end user from the web UI

Technologies:

Java, FA, OSC/ERP/HCM, WSDL/SOAP, REST, JSON, XSI, XML Schema, XDK, JAXP, WSDL4J, DOM, WS-Security, Jersey REST/JSON, MVC, ADF, ADE, Maven, Junit, XMLUnit, SoapUI,

Leadership:

- Architected a business event connector framework, enabling real-time integration for enterprise applications.
- Drove the adoption of event-driven patterns, empowering teams to build scalable, asynchronous integrations.
- Mentored developers on advanced payload manipulation and schema evolution.

7. Oct. 2010 – Apr. 2014, Senior Cloud DevOps/Infrastructure Automation Engineer, Comcast T&P , Philadelphia, PA,USA

As a senior engineer in BITT(Build Integration Test Team), provided DevOps support for multiple development teams and supporting the developers for SDLC infrastructure system design/implementation, and Build/Release/Deployment/SCM/CI automation., designed, developed and implemented the AutoBuild system, OpenSource performance monitoring system for the SDLC systems, two steps bare-metal Linux bootstrap system for HP servers, Linux HA Clusters for SCM/Maven/CI/Crowd servers, Ruby/Rails software system maintenance and improvement. Managed the SCM(Gerrit/Subversion), CI(Bamboo/Jenkins) and maven repo (Nexus and Artifactory) system. Designed and developed automation scripts (bash/ruby) to improve work efficiency. Led and Trained junior team members.

7.1 AutoBuild System for Build/Release more than 200 Microservice projects with GIT/SVN/Maven

Developed and automated the AutoBuild system to streamline build, release, and dependency management for over 200 microservice projects, reducing release cycles from days to hours.

Highlights:

- Design and implement the error handling and logging system for ruby bash commands
- Implement build dependency graph (directed acyclic graph - DAG) using RGL for more than 200 projects
- Refactoring autoBuild to adapt to much large organization development practices
- Design and implement the whole AutoBuild system
- Using ruby and bash scripts wraps multiple manual steps to automate the whole branching and tagging process
- Modify maven versions plugin to fix the bug which cannot update the dependency version property variables and the bug which cannot update the latest snapshot version to the latest released version
- Detect and update the dependency versions in pom.xml automatically using maven versions plugin and maven repo servers REST API
- Design and implement the branching/tagging process
- Developed ruby/bash scripts to lock/unlock remote GIT branch in Gerrit server during branching and cutting tags period
- Design code standards and rule to follow AutoBuild conventions for developers
- Applied convention over configuration for AutoBuild system
- Develop ruby scripts to post json message to a BOM management system to allow the artifacts ready to deploy
- Develop automation script to post message to Jira and Confluence page using Atlassian Jira-cli and Confluence-cli
- Develop bamboo scripts using bamboo command-cli to create plans from git/svn repos automatically
- Install and configure Bamboo, Nexus, Artifactory, Crowd/OpenID, Gerrit and Subversion servers.
- Maintain Ruby/Rails BOM management system
- Setup maven release, versions, toolchains, dependencies ...etc. plugins

Technologies:

Ruby/Rails, gems, Bash, Maven, GIT, SVN, Bamboo/Jira/Confluence CLI, Nexus/Artifactory REST/XML, SDLC, CI/CD, Gerrit/Subversion, MySQL/PostgreSQL, Crowd/OpenID, Capistrano,

Leadership:

- Invented and led the development of the AutoBuild system, reducing release cycle time from days to hours.
- Established code standards and conventions for automated build and release processes.
- Enabled organization-wide adoption of automated dependency management and versioning.

7.2 Configuration management and deployment automation with Puppet and Capistrano

Implemented configuration management and deployment automation using Puppet and Capistrano, eliminating manual errors and accelerating application delivery across environments.

Highlights:

- Setup puppet master server and agent in Linux Centos for DevOps and CI/CD environments
- Designed and implemented develop/build/deployment process for the puppet templates files by using GIT
- Developed puppet templates file for linux os patch and lib yum installation and updates
- Developed capistrano deployment automation tasks and functions used by operation team for production deployment
- Maintained and troubleshooting complex configuration and deployment automation issue within Capistrano and Puppet
- Developed puppet templates and capistrano tasks to automate the configuration and deployment process for Microservice with Dev/CI/CD environments
- Developed Capistrano tasks to automate the Liferay deployment to Dev/QA/Staging/Prod environments with JDBC driver configuration to Oracle DB Server 11g

Technologies:

Puppet, Capistrano, Ruby, bash, CI/CD, GIT, Maven, Perl, Python,

Leadership:

- Introduced configuration management automation, eliminating manual errors in deployment.
- Developed reusable deployment tasks, empowering operations teams to manage complex environments.
- Provided training and documentation for DevOps best practices.

7.3 Continuous Integration (CI) and Continuous Deployment (CD) with Bamboo and Jenkins

Standardized and automated CI/CD pipelines for 200+ microservices using Bamboo and Jenkins, improving release reliability and developer productivity.

Highlights:

- Designed and implemented the CI/CD clustering environments using Bamboo and Jenkins for all different development and release branches
- Developed automation scripts using Ruby/Bash/Perl/Python to launch the CI/CD environments with complex dependencies for more than 200 Microservice
- Developed automated integration scripts to integrated Bamboo/Jenkins with Jira/Confluence, GIT/Maven repo
- Designed and implemented complex tasks/jobs chains/pipeline with trigger or events
- Maintained and Troubleshooted complex CI/CD issues within the clustering environments with multiple Subsystems for more than 200 Microservice
- Developed automation scripts to automatically create Bamboo tasks from git repo and capistrano
- Developed Capistrano automation tasks to deploy Microservice stack with multiple Microservice to Dev/QA/Staging/Prod
- Developed Capistrano tasks to generate configuration file for HAProxy automatically
- Setup Bamboo and Jenkins Servers Cluster with multiple remote agents/nodes as the build/deployment servers

Technologies:

CI/CD, Bamboo, Jenkins, Ruby, Bash, Python, GIT, Maven, Jira/Confluence,

Leadership:

- Standardized CI/CD pipelines across 200+ microservices, improving release consistency and reducing deployment errors.
- Automated integration between build systems and project management tools, streamlining developer workflows.
- Mentored junior engineers in best practices for automation and pipeline design.

7.4 Performance Monitoring System for Jetty, JVM, Linux and Splunk

Designed and integrated a comprehensive performance monitoring system for JVM, Linux, and Splunk, enabling proactive detection and resolution of system bottlenecks.

Highlights:

- Designed the whole monitoring system using open source projects such as graphite, gdash, metricsd, stated, jmxtrans and god...etc.
- Developed UDP metrics connectors and ruby splunk searcher to collect splunk stats
- Developed UDP connection for open source project jmxtrans, metricsd by using Java and Scala
- Developed Capistrano task to dynamically generate gdash templates for the metrics graphs and jmxtrans configuration files.
- Developed scripts manage the ruby processes by using open source God project
- Fixed the bug cannot listing the jmx attributes in details for open source project jmxsh

Technologies:

Metrics, graphite, jmx, splunk, Ruby, splunk-ruby-SDK, performance monitoring, UDP, Capistrano, god, bash, gdash, graphite, statsd, metricsd,

Leadership:

- Unified disparate open-source tools into a single, cohesive monitoring platform, enabling proactive performance management.
- Led the integration of custom metrics connectors, providing deep visibility into JVM and OS health.
- Mentored team members on advanced monitoring and troubleshooting techniques.

7.5 Secured FTP server with virtual users and admins users with complex access permissions

Engineered a secure FTP/SFTP environment with advanced user and group access controls, automating user management and improving security compliance.

Highlights:

- Setup vsftp server with virtual ftp only user which can only do ftp access and do not have local account associated
- Setup sftp server with admin users which are chrooted such that they do not have access to the files
- Setup the hybrid users with both ftp and sftp access
- Modified open source vuser administration bash scripts to adapt to the special requirements
- Developed script to sync the virtual user and local user for the hybrid user account
- Setup complex user and group permission to do access control for different users

Technologies:

ftp, sftp, chroot, bash, linux, user/group unix permission, sshd,

Leadership:

- Engineered a secure, multi-user FTP/SFTP environment, balancing usability with strict access controls.
- Automated user and group management, reducing administrative overhead and improving security compliance.
- Documented complex permission structures for future maintainers.

7.6 BOM (Bill Of Materials) Administration and Management Application (BAMA) Enhancement and maintenance

Enhanced and maintained the BAMA system for automated deployment and management of microservices, introducing customizable deployment features and integration with CI/CD tools.

Highlights:

- Installed and configured Ruby on Rails 3 for BAMA for different environments with Delayed_jobs, Websocket server
- Developed the Enhancement feature for BAMA to enable customization of list of Microservice deployment to different systems
- Developed Rails3 ActiveRecord DB Migration scripts to add new columns to the DB tables
- Developed Rail3 Model, View, Controller, Routes, Websocket and Validation jobs
- Developed Websocket server customized startup script
- Developed scripts to integrate Capistrano and BOM GIT repo and BAMA together to deploy Microservice with matching versions for Dev/QA/Staging/Prod environments
- Developed automation ruby scripts to post JSON message to BAMA for the released artifacts
- Developed scripts to upload artifacts to Maven Repo (Artifactory and Nexus) by using their REST APIs
- Setup SSO (Single-Sign-On) for BAMA with Atlassian Crowd ID and LDAP-based Authentication

Technologies:

Ruby, RoR 3, Rails 3, ActiveRecord, ActionView, ActionController, ActionMailer, Websocket, linux, JSON, Capistrano,

Leadership:

- Enhanced deployment automation for 200+ microservices, streamlining environment management.
- Introduced customizable deployment features, empowering users to tailor release processes.
- Provided cross-functional training on Ruby on Rails and deployment automation.

7.7 Two Steps bare-metal Linux bootstrap system for HP Servers

Developed a two-phase, fully automated Linux server bootstrap system for HP hardware, enabling zero-touch provisioning and configuration.

Highlights:

- Design the two steps bootstrap system using tftp/pxe/dhcp and cobbler
- Statically build netstat tool using make/gcc to run it in HP-Toolkit with limited dynamic libraries
- Developed a XMLRPC bash client to remotely connect back to the DHCP/cobbler server to add a new system from the bare-metal box during the first boot
- Modify the bootstrap script for the first boot to collect the mac-address and parse a boot configure file to add a new system to cobbler server.
- Physically upgrade the memory, disks and fusion IO adapter for HP blade servers
- Setup ILO and IPMI
- Configure disk-array and virtual storage management (lvm, pv and vg..etc.)

Technologies:

TFTP, PXE, DHCP, DNS, cobbler, RPC, xml, bash, ILO 3, IPMI, LVM, HP Servers, CentOS 6,

Leadership:

- Designed a two-phase, fully automated server bootstrap process, reducing manual provisioning time.
- Developed custom scripts for dynamic system registration and configuration, enabling true zero-touch deployment.
- Trained operations staff on advanced PXE, DHCP, and cobbler automation.

7.8 Linux HA for Git Repo(Gerrit) and Maven Repo (Artifactory/Nexus) Servers

The two nodes Linux HA cluster is for Git repo (Gerrit) / Maven Repo(Artifactory and Nexus) master/slave failover system. Using pacemaker/corosync/cman or keepalived/haproxy, it setup Gerrit, artifactory/Nexus master/slave automatically failover cluster with PostgreSQL/RAC and GFS2 over SAN / DRBD as backend DBMS and distributed file system.

Highlights:

- Design Linux HA master/slave failover cluster for Gerrit, Artifactory.
- Configure pacemaker/corosync/cman for two-nodes cluster
- Create Pacemaker resources and constraints for the services using pcs
- Setup and configure keepalived and haproxy for Nexus, artifactory
- Developed HAProxy initd scripts to automatically failover to the latest good configure file if capistrano generate incorrect config file
- Setup Distributed filesystem GFS2 over SAN(Multipath) and DRBD
- Developed keepalived check scripts for Nexus and artifactory
- Setup PostgreSQL master/slave cluster over SAN

- Setup LVM over SAN
- Setup OpenID authentication for Gerrit
- Setup Gerrit cluster over SAN and PostgreSQL
- Setup Artifactory Cluster over SAN and RAC
- Setup Nexus Cluster via replication scripts with rsync
- Setup Gerrit OpenID authentication with Atlassian crowd backend with Multiple LDAP servers

Technologies:

Pacemaker, Corosync, Cman, Keepalived, HAProxy, GFS2, DRBD, SAN, Gerrit, Artifactory, Nexus, PostgreSQL, OpenID, LDAP, Crowd,

Leadership:

- Architected a high-availability cluster for critical SCM and artifact repositories, ensuring business continuity.
- Automated failover and recovery processes, minimizing downtime during incidents.
- Standardized HA deployment patterns for future infrastructure projects.

7.9 Automation scripts and tools for DevOps

Created a suite of automation tools and scripts to streamline DevOps workflows, improve efficiency, and reduce manual toil across the SDLC.

Highlights:

- Develop migration tools to migrate git repos from different remote git repos(Gerrit, stash and gittorious) to new Gerrit server
- Develop scripts to find the largest commit object and rewrite git history to improve git performance
- Develop Gerrit/git hooks for rally and jira
- Develop multiple automation tools using confluence, jira and bamboo CLI
- Developed tools to automatically build the source dependency tree and branch/tagging parallel
- Maintain configuration management tool Capistrano and automate the environment and service configuration
- Using Puppet to automate the system level configuration
- Developed Nexus replication scripts from master to slave automatically with keepalived
- Developed maven circular dependency checking scripts and created nightly CI jobs to run it against more than 100 Microservice projects
- Developed Capistrano logic to automatically check ruby version and install the right ruby version with rbenv
- Developed bash scripts to launch python scripts to generate data parity reports and upload to confluence pages automatically
- Developed automation tools to create Rally defect ticket and upload the heapDump file and grep the exception logs
- Migrated Solr from Single-core to multi-core with automated Capistrano tasks
- Developed Capistrano task to automatically migrate the Solr data while upgrade jetty to new version
- Installed and configured Python 2 and Python 3 in the same linux host
- Developed Capistrano logic to automatically detect DB schema changes made by developers from source code

Technologies:

Ruby, Bash, Python, Capistrano, Jira/Rally, Bamboo/Confluence, Git/Gerrit, puppet, Nexus replication, REST/XML,

Leadership:

- Developed a suite of automation tools, significantly improving DevOps efficiency and reducing manual toil.
- Pioneered migration and optimization scripts for large-scale git repositories.
- Shared best practices and tools with the wider engineering community.

7.10 CoreOS and Docker prototyping and testing

Led the design and implementation of dynamic development and integration test environments using CoreOS and Docker, reducing setup time and increasing productivity.

Highlights:

- Helped design and implement the docker LXC system
- Setup CoreOS stateful and stateless using pxe image via cobbler and tftpboot/DHCP
- Create Docker images for CentOS/Java/Ruby/Haproxy
- Create ruby Capistrano tasks to dynamically provision docker instances with haproxy as the internal LB on demand
- Automate the dynamic deployment and provisioning process by using docker rest API and tools

Technologies:

CoreOS, Docker, Ruby, REST, LXC, cobbler, pxe, tftpboot, DHCP,

Leadership:

- Led the design and implementation of a dynamic development and integration test environment using Docker, significantly reducing setup time and improving developer productivity.
- Created reusable Docker orchestration templates, enabling rapid prototyping for SDK and connector teams.
- Trained team members on advanced Docker networking and orchestration concepts.

7.11 SDLC process management and junior team member training and mentoring

Managed SDLC processes and mentored junior engineers, standardizing build, release, and deployment practices for over 200 projects.

Highlights:

- Help to design and identify the build/release/deploy and configuration management strategies
- Design standard development and code standards for developers to follow the autoBuild process.
- Using Capistrano to manage deployment configuration for different environments(Dev, QA, Staging and Prod..etc.)
- Communicate with developers to figure out the build/release issues
- Troubleshooting build/release and configuration issues
- Lead small team to develop autoBuild system using agile methodology
- Train and guide new team member
- Write confluence pages for documents
- Presents to other team members in brown bag session

- Transfer knowledge to other team member
- Administration the SCM/CI/Maven Repos clustering environments and SDLC tools
- Research and design github enterprise branch/release process
- Develop scripts to build dependency graphs and design algorithm to simplify it to dependency tree for more than 150 projects (git repos)

Technologies:

SDLC, SCM, Git/Subversion, GIT Hooks, Ruby, Bash, Confluence, Jira, Agile,

Leadership:

- Managed the SDLC process to build/release/deploy more than 200 projects, maintaining maven repo servers, CI (Bamboo), SCM (Subversion and Git/Gerrit) and clustering integration environments.
- Identified and troubleshooted build/release/deploy issues, and trained and mentored junior team members.
- Wrote confluence pages for documentation and presented to other team members in brown bag sessions.

8. Nov., 2006 – Oct., 2010, Senior Software Engineer, Boomi Inc. , Berwyn, PA,USA

As the market and technology leader in on-demand integration, Boomi is the industry's first and leading integration platform-as-a-service and a two-time SIIA CODiE award winner - Best Application Integration Solution (2010) and Best On-Demand Platform (2009).

8.1 Boomi On-Demand Business Integration Cloud Service

Designed and developed a scalable, on-demand SOA integration platform enabling visual business process design, automated deployment, and seamless integration with desktop, ERP, and SaaS systems.

Highlights:

- Design and implement synchronous/asynchronous message framework using XML/JAXB/REST/HTTP
- Design and implement dynamic reflection invocation connector framework using dynamic proxy and customized class loading to failover from local file system to remote webserver
- User Management Subsystem (PKI certificate token based authentication and authorization)
- Design and implement connectors to desktop accounting systems (QuickBooks, MAS 90, MS Dynamics Great Plains, Peachtree), ERP systems (MAS 500, SAP, MS Dynamics Navision, MS Dynamics CRM on-premise 4.0 and Online) and SaaS system (Parature, AutoTask).
- Design and implement Java to COM/DCOM/COM+/ .NET/WCF integration subsystem
- Migrating data processing, mapping function …etc. modules from previous non-SOA version
- Design and develop rich internet UI using openlaszlo
- Design and deploy the load balancing and clustering SOA system in OpSource and Rackspace data center
- Deploy SOA system in Amazon's EC2 and S3
- Multiple threads concurrent subsystem design and implementation
- Design and implement distributed cache in cluster using ehcache
- Develop, build, deploy and release projects using IDEA, maven, subversion and TeamCity
- Tracking bugs/Issues and interacting with QA/Support team using Jira Enterprise
- Using confluence page to document ideas and functional design
- Participate in agile development with scrum framework

Technologies:

Java, XML, XSD, JAXB, REST, HTTP, Hibernate, openlaszlo, SSL, PKI, com4j, .NET 3.0/2.0/1.1, COM/DCOM, Linux, QBXML, eConnect, servlet, Jetty, IDEA, Maven, subversion, Confluence, Jira, MySQL, RAID, SQL server, QuickBooks 2005, Sage MAS 90/200/500, MS Dynamics GP 9.0/10.0/2010, SOA,

8.2 Cloud Platform/Infrastructure Design and Implementation for Boomi On Demand Production Environment

Architected and managed a distributed, high-availability production environment for Boomi's cloud platform, including clustering, monitoring, and migration of critical infrastructure.

Highlights:

- Plan, Design and setup the clustering SOA production environment in OpSource and Rackspace
- Maintain, build and release the QA/production environment.
- Migration of the production environment, application and MySQL DB data from OpSource to Rackspace
- Setup and configure Apache Solr searching and indexing server
- Create cron jobs and shell scripts to monitor hosted AtomSphere environment
- Setup newrelic monitoring environment
- Research and proto-type cluster solutions for the large SOA system.
- Design distributed cache system using ehcache to reduce the DB access traffic from front end.
- Design and implement the dynamic http polling algorithm on the front end to reduce the possibility of the suppress storm requests.
- Using comet, jetty continuation to design an asynchronous http push (long polling) events system.
- Using worker-stealing algorithm to design and implement an Apache solr indexing cluster with large throughput transactions (millions per day)
- Research and proto-type MySQL clustering technologies
- Setup and configuration Jetty and Openlaszlo clustering system (NFS sharing)
- Communicate with Rackspace support engineers with their tickets system to tracking issues
- Design, implement and troubleshooting Apache solr out of memory and DB table transaction lock and response time vibration problem.
- Using Sun JVM troubleshooting utilities (visualVM, jmap, jstat,..etc.) to investigate the heap new generation and old generation usage
- Tuning heap size and GC algorithm parameters to solve the out of memory issue
- Collect response time statistics data to diagnose software system vibration
- Using Unix vmstat, top,...etc. to investigate the File system IO, CPU usage

Technologies:

64-bits Redhat Enterprise Linux, MySQL clustering, Apache, PHP, Jetty, 64 bits Java, Apache Lucene/ Solr, Openlaszlo, distributed cache, distributed and parallel algorithm, ehcache, comet, continuation, DWR, JVM tuning, jmap, jstat, Sun JVM Heap, GC, vmstat,

8.3 Desktop Accounting and ERP Enterprise Applications Integration (EAI)

Engineered integration solutions for leading desktop accounting and ERP systems, enabling automated business data exchange and process automation.

Highlights:

- Integrate QuickBooks (US, UK, Canada 2002-2008,) and QuickBooks Online Edition using the QBXML request processor.
- Integrate Microsoft Dynamics Great Plains (9.x, 10.x and 2010) using eConnect (9.x, 10.x and 11) and WCF.
- Integrate Microsoft Dynamics CRM on premise (3.0 and 4.0) and online using Web Services with NTLM authentication and windows live ID authentication
- Develop integration solution for Sage Peachtree Accounting system using its Import/Export engine API
- Integrate with ERP (SAP, Oracle e-Business, PeopleSoft, JD Edwards and Siebel) using netmanage's Librados API.
- Setup QuickBooks, Sage Peachtree, MS Dynamics GP desktop accounting system for testing
- Setup Dynamics CRM 4.0 in EC2 windows server 2008 domain environment for testing
- Research integration with QuickBooks POS system
- Research integration with Sage Software (MAS 90/200, 500 ERP, Simply, Accpac ERP).
- Research integration solution with Microsoft Dynamics Navision 4.0, 5.0 and 2009 using XMLPort, .NET Web services and Communication Components with windows socket bus adapter.

Technologies:

QBXML RequestProcessor, Microsoft eConnect 9.x/10.x/11, Librados, SAP R3, BAPI, REST/XML, COM/COM+/DCOM, .NET, MS Dynamics Navision, MS Dynamics CRM, SOAP/WSDL, WCF, XMLPort, C/AL, CodeUnits,

Leadership:

- Led the integration of multiple accounting and ERP systems, enabling seamless business data exchange.
- Researched and implemented SDK-based solutions, setting a foundation for future integrations.
- Documented integration strategies for cross-team knowledge sharing.

8.4 Amazon EC2 and S3 SOA HA Environment

Built and maintained a high-availability SOA environment on AWS EC2 and S3, implementing clustering, load balancing, and automated backup for robust cloud operations.

Highlights:

- Install and setup the Amazon EC2 client security (PKI token based authentication), commands and AMI tools
- Setup Amazon EC2 large/small Linux/windows server instance with static IP address and EBS.
- Install and configure two Jetty instances (Web UI + App server) in EC2 instance (Fedora 7)
- Setup Apache httpd load balancer (mod_proxy_ajp and mod_proxy_balancer) for jetty 6.1.x
- Setup NFS server and client to share data between two EC2 instances
- Install MySQL 5.x server, create DB user, grant privileges and replicate the data.
- Install and configure Solr in Jetty 6.x and create startup script
- Create Jetty startup script to create log directory and grant permissions to the running users.
- AMI bundles and images maintenance (create AMI bundles and upload it to S3, register AMI, remove AMI from S3).
- Rescue the broken AMI (Download AMI bundles from S3, unbundle and mount it as loop device).
- Setup EC2 security group and firewall.
- Troubleshooting EC2 instance network and AMI booting problems

- Troubleshooting timeout and performance problem
- Setup Fedora 6/7/8 x86_64 EC2 large instance with Apache, MySQL
- Troubleshooting MySQL 5.x 64-bit access denied problem
- Troubleshooting MySQL 5.x performance (table lock timeout and setup the right isolation level)
- Configure distributed cache for the HA cluster using ehcache
- Setup EBS for the HA cluster
- Setup Rightscale auto-deploy scripts to start EC2 instance with Boomi AtomSphere built-in.
- Setup Windows Server 2008 domain environment with SQL Server
- Implement cron job and shell scripts to backup EC2 instance data and EBS snapshot to S3

Technologies:
Fedora 6//7/8, AWS, EC2, S3, Jetty 6.x/7.x, MySQL 5.x, Apache httpd 2.x and Solr 1.3.x/1.4.x, NFS, shell script(bash and sh), ehCache, cron,

Leadership:
- Engineered a high-availability SOA environment on AWS, pioneering cloud adoption within the team.
- Automated deployment and backup processes, ensuring data integrity and system resilience.
- Provided training on AWS best practices and troubleshooting.

8.5 Setup Development, Build and Deployment Environment for QA, Staging and Production Environment

Established and automated development, build, and deployment environments for QA, staging, and production, streamlining release management and system administration.

Highlights:
- Install and configure java, maven and subversion.
- Install and configure MySQL 5.x, PHP5.x, Jetty 6.x, openlaszlo, quartz and Apache solr.
- MySQL 5.x administration (create database, create user, grant permission and populate data)
- Create Jetty startup script and DB update SQL script
- Upgrade MySQL DB Structure
- Using TeamCity create deploy and release projects
- Create PHP scripts connecting to MySQL DB to maintain user accounts
- Create unix shell scripts and Apache ant xml to release to QA, Staging and Production environment

Technologies:
Maven 2.x, subversion, TeamCity, PHP 5.x, Apache Http server 2.3, Apache Ant, MySQL5.x, Jetty 6.x, Openlaszlo 3.4, Apache Solr, Fedora 8,

Leadership:
- Standardized build and deployment environments across all stages, improving release reliability.
- Automated user account and database management, reducing manual errors.
- Mentored team members on build automation and environment setup.

8.6 Research virtualization technology and implement Virtual Machines using VMware server and workstation

Researched and implemented virtualization solutions using VMware, optimizing resource utilization and performance for development and testing environments.

Highlights:
- Research, compare and evaluate VMware workstation 6.2, VMware server 1.4 and Microsoft Virtual PC.
- Install and Configure VMware Workstation 6.2, VMware Server 1.4, VMware Server Console.
- Adding new hard disk to Logic Volume Group (pvcreat, vgextend, lvcreate,mke2fs)
- Using top, vmstat, iostat and sar and some benchmark tools(dbench, unicbench) to diagnostics the Linux system performance
- Upgrade Dell and LSI RAID Controller BIOS firmware and Seagate SCSI Hard disk driver
- Tuning Disk I/O performance via distributing the VM disk to different partition
- Install and setup VNC server and client on Linux
- Install and configure subversion 1.4, Maven 2.x, PHP5.x, Apache Httpd 2.x, MySQL 5.x.

Technologies:
Virtualization, paravirtualization, VMware Workstation 6.2 and VMware Server 1.4, Xen, Microsoft Virtual PC, Fedora 8, RAID, SMP, Performance tuning, Subversion, Maven, PHP, MySQL,

Leadership:
- Evaluated and implemented virtualization solutions, enabling efficient resource utilization.
- Developed performance tuning guidelines for virtualized environments.
- Shared virtualization best practices with the broader engineering team.

8.7 Java to C/C++, DLL, COM/COM+/DCOM/.NET Integration

Developed seamless integration between Java applications and native Windows technologies (C/C++, DLL, COM/DCOM, .NET), enabling cross-platform interoperability.

Highlights:
- Research and evaluate open source technologies such as com4j, JNBridge, ikvm, etc. for java-com and java-.net integration
- Develop prototype application using the Microsoft Visual Studio 6.0, 2003 and 2005 (C# and VB)
- Configure COM/COM+/DCOM/.NET run time environment (regsvr32, dcomcnfg, gacutil, regasm, ..etc.)
- Develop java to win32 DLL application using JNA (Java Native Access)
- Develop java to COM/COM+/DCOM object application using com4j and register com object programmatically.
- Using windows ComAdmin Objects access COM+ catalog to install and configure .NET assemblies' security permission programmatically in C#.
- Using iKVM to integrate Java with .NET run-time
- Using .NET System.Reflection namespace to dynamically invoke ComAdmin objects methods
- Using open source JNA (Java Native Access) connect to win32 DLLs from java

Technologies:
Java, .NET 3.0/2.0/1.1, COM/DCOM, Com4j, iKVM, C#, VB.NET, JNI, JNA, Win32 DLL, COMAdmin,
Leadership:
- Pioneered seamless integration between Java and native Windows technologies, enabling cross-platform interoperability.
- Developed proof-of-concept applications that set technical direction for future integration projects.
- Provided technical guidance on COM/DCOM and .NET integration to the engineering team.

8.8 SaaS system integration
Designed and delivered integration solutions for leading SaaS platforms using REST/XML and SOAP/WSDL, expanding product connectivity and automation.

Highlights:
- Integrate with Parature.com using its' REST style API
- Integrate with AutoTask using SOAP/WSDL
- Integrate with Zuora using SOAP/WSDL

Technologies:
REST/XML, SOAP/WSDL, namespace, DOM, soapUI,
Leadership:
- Designed and implemented robust integration solutions for leading SaaS platforms, expanding product connectivity.
- Standardized REST and SOAP integration patterns, improving maintainability and scalability.
- Mentored junior engineers on SaaS API best practices.

8.9 SAP system integration
Engineered robust integration between Java and SAP systems using web services and SAP JCO, enabling automated business process integration.

Highlights:
- Install and configure SAP NetWeaver 2006s/2007 test drive on Linux environment
- Research and Prototype different SAP system integration solutions
- Create and publish web service (SOAP/WSDL) for RFC and BAPI in SAP system using ABAP workbench
- Self-teaching SAP basis system in two weeks
- Troubleshooting librados/sapjco issues on SAP NetWeaver
- Design and Implement Web Services (SOAP/WSDL) solution to integrate Java to SAP RFCs
- Administrator SAP ABAP web services environment using SAP front-end UI.

Technologies:
SAP R3, NetWeaver, Web Services, SOAP, WSDL, wsdl4j, SAP-JCO, RFC, BAPI, ABAP, Java,
Leadership:
- Pioneered integration between Java and SAP systems, enabling seamless business process automation.
- Self-taught SAP basis and led troubleshooting efforts, accelerating project delivery.
- Documented integration patterns for future SAP-related projects.

9. Jun., 2006 – Nov.,2006, Senior System Engineer Manager, Vantage learning , Newtown, PA,USA
As the leader in cost-effective, high-volume, secure, scalable online assessment and instructional programs for K-12 and higher education, Vantage Learning leverages technology such as artificial intelligence, natural language understanding, and web-based learning objects. Vantage Learning has received accolades ranging from the prestigious CODIE Award for best instructional technology.

9.1 Large scale SOA e-learning system architecture design and blueprinting
Architected a scalable SOA-based e-learning platform leveraging Java/.NET and clustered PostgreSQL/MySQL databases, enabling secure, high-volume online assessment and instruction.

Highlights:
- Researching and blueprinting SOA new technologies (WCF, WPF, XAML, AJAX, XML, SOAP, WSDL)
- Setup proto-type environment for distributed session in a clustering SOA system
- Design large-scale SOA clustering load balancing system
- Design integration solution for Java and .NET
- Researching on DB clustering environment for PostgreSQL and MySQL

Technologies:
SOA, E-Learning, Struts, J2EE, .NET 2.0/3.0, Resin, SOAP, WSDL, UDDI, XML, PostgreSQL, MySQL,
Leadership:
- Blueprinted a scalable SOA e-learning architecture, setting the technical vision for the organization.
- Researched and introduced new SOA technologies, driving innovation in the e-learning platform.
- Facilitated cross-team collaboration between Java and .NET development groups.

9.2 Large Data Center Design and Administration
Designed and managed a mission-critical data center with 100+ servers and enterprise network devices, supporting 30+ applications and millions of user connections with high availability.

Highlights:
- Design data center network architecture and plan budgets on new devices and spare parts
- Design and implement clustering and load balancing hosting environment
- Design and deploy large-scale SOA clustering system
- Deployment of clustering hosting environment with enterprise DBMS
- RAID 1,0,5 and SAN design and setup

- Veritas backup system setup and maintenance
- Unix and DBMS maintenance
- Web based system troubleshooting and debug

Technologies:
Solaris 5,7, Linux(Redhat,Fedora), F5 Big-IP load balancers, Switch, Router, VLAN, UPS, RAID 1,0,5, Windows Server, Terminal server, VNC, SPARCStorage, Tape Drive, Veritas, Apache, Resin, PostgreSQL,

Leadership:
- Led the design and deployment of a mission-critical data center serving millions of connections.
- Standardized clustering and load balancing practices, improving system reliability.
- Mentored junior engineers in data center operations and troubleshooting.

9.3 Sun SPARCStorage SSA Array 100 and Sybase SQL Server (10.x, 11.x) Data (DB schema, data and Stored-Procedure) Recovery

Recovered critical patent data and database assets from failed Sun SPARCStorage SSA Array and Sybase SQL Server, ensuring business continuity and data integrity.

Highlights:
- Troubleshooting the hardware failure of Sun SPARCStorage SSA Array
- Install and Configure Sybase SQL Server
- Use bcp, dump and load database to recover the crashed databases and stored procedure
- Output the SQL statements (DDL, DML) of the old data

Technologies:
Solaris 5,7, Linux(Redhat, Fedora), Sybase SQL Server 10.x, Sybase ASE 11.x, 12,x 15.x, SSH, SPARCStorage SSA Array 100, Tape Drive,

Leadership:
- Directed the recovery of critical patent data, ensuring business continuity after hardware failure.
- Developed and documented robust data recovery procedures for future incidents.
- Coordinated efforts between hardware, database, and software teams.

9.4 Ecommerce Open Source Software installation and Configuration

Deployed and configured open source e-commerce platforms (osCommerce, x-cart, Sales-n-Stats) on PHP/MySQL, standardizing installation for rapid business enablement.

Highlights:
- Install and Configure test environment for osCommerce with CREloaded on PHP, MySQL environment
- Install and Configure x-cart with Sales-n-Stats module on PHP, MySQL environment

Technologies:
Linux(Redhat, Fedora), PHP5.x, MySQL 5.x, Apache 2.x,

Leadership:
- Led the evaluation and deployment of open source e-commerce solutions, expanding business capabilities.
- Standardized installation and configuration processes for rapid future deployments.
- Provided training on PHP/MySQL stack to team members.

9.5 Resin Application Server Distributed Session Design and Setup

Implemented distributed session management and HTTP load balancing on Resin Application Server, enhancing scalability and reliability for enterprise web applications.

Highlights:
- Setup resin cluster environment with distributed session enabled
- Setup http load balancer for resin with sticky bit

Technologies:
Linux(Redhat, Fedora), resin 3.0, Java, Servlet, distributed session, cluster, load balancer (software and hardware),

Leadership:
- Designed and implemented distributed session management, improving application scalability.
- Automated cluster and load balancer setup, reducing manual configuration time.
- Documented best practices for session management in clustered environments.

10. May, 2002 – Jun.,22,2006, Senior Programmer/Analyst and Unix System Administrator, Computer Science, Texas Tech University , Lubbock, Texas,USA

System Administrator for networked Unix(Solaris, AIX, Mac OSX and Linux) and windows servers with SSO for Kerberos and LDAP, Veritas Netbackup for Disk Array with RAID 0/1/5, Migration from NIS/NIS+ to LDAP and Postfix/Sendmail conifiguration for the research and teaching lab for computer science department.

10.1 Unix network System and Software routine maintenance

Performed daily administration and maintenance of Solaris, AIX, and Mac OSX network systems, ensuring reliability and security across research and teaching labs.

Highlights:
- Troubleshooting Unix network system with multiple services (NFS, NIS+, SMTP, LDAP ...etc.) running
- Developed system maintenance scripts (Perl, bash)
- Unix Security design and analysis
- Compiled, Installed and configured system and development software
- NIS+ user accounts managements

- Diagnosed hardware and software problems

Technologies:
Solaris 8,9,10, Linux, AIX, NFS, NIS/NIS+, LDAP, SMTP, IMAP, POP3, WebMail, SSH, Samba, Kerberos, Netgroup, Jumpstart, User Accounts, Quota, License server, GNU Make, GCC, Java, RAID, Tape Library, Veritas, Apache, MySQL, Sun Ray Server, PAM, Security, Syslog,

Leadership:
- Standardized system maintenance procedures, improving uptime and reliability across multiple platforms.
- Developed and shared custom scripts for automation, reducing manual workload.
- Mentored student assistants in Unix system administration.

10.2 Setup Veritas Netbackup System with RAID0/1/5 and Tape Library

Implemented enterprise backup and recovery solutions for large RAID storage arrays and virtual file systems using Veritas NetBackup and tape libraries.

Highlights:
- Setup Veritas NetBackup Enterprise Server 5.1 on Solaris 9
- Setup Veritas NetBackup client software on Linux and Solaris
- Designed and setup tape backup policies
- Installed and setup Sun StorEdge Tape Library L25 and L9
- Installed and setup Sun StorEdge A100 and 3300
- Configured hardware and software RAID 0,1 and RAID 5
- Setup Logical Partitions on RAID systems
- Setup volume and file system on the RAID system
- Maintained (expand partition, clean file system) a large virtual file systems. (>300GB).
- Backup and restored the system and user data using Veritas software

Technologies:
Veritas NetBackup Enterprise Server 5.1, Veritas NetBackup Client, RAID 0,1,5, SCSI, Volume Manager, VFS, Tape Library,

Leadership:
- Led the implementation of enterprise backup solutions, safeguarding critical research and teaching data.
- Developed backup and recovery policies, ensuring data integrity.
- Trained staff and students on backup best practices.

10.3 Setup LDAP, Kerberos KDC and Cross-Realm system

Designed and deployed a secure Single Sign-On (SSO) system with LDAP, Kerberos KDC, and cross-realm authentication, unifying access across Unix and Windows platforms.

Highlights:
- Setup SunOne Directory Server 5.2 and populate the LDAP data from NIS+ tables
- Setup SEAM Kerberos KDC, cross-realm authentication with Microsoft Active Directory Server
- Configure Kerberos and LDAP client and PAM authentication
- Configure SASL/GSSAPI authentication to LDAP server and identities mapping between SASL principal

Technologies:
SASL-2.1, GSSAPI, SEAM 1.2, SunOne Directory Server 5.2, NIS+, LDAP V3, PAM, Solaris 9, Windows Server 2003, Active Directory,

Leadership:
- Designed and implemented a secure, unified authentication system for the department.
- Coordinated cross-platform integration between Unix and Windows environments.
- Documented SSO setup and provided training for system administrators.

10.4 Migrated email system from UNIX mail server to Windows Exchange Server

Led the migration of departmental email from Unix servers to Windows Exchange, configuring secure SMTP/IMAP/POP3 with SSL for diverse clients.

Highlights:
- Led the email migration team and plan the email migration for whole department (more than 50 faculty/staff and more than 500 students email accounts)
- Configured Unix/windows mail client to connect to Microsoft exchange server via IMAP with SSL authentication
- Written reports and provide recommendations to Department Computer Use Committee
- Implemented IMPA/S mail client to Microsoft Exchange Server

Technologies:
Sendmail, Postfix, Unix Mail Client, Outlook, Solaris 9, Linux, Windows Server, Exchange Server, IMAP/S, POP3/S, SSL,

10.5 Secured Postfix SMTP server with SASL Authentication and Verification

Configured Postfix SMTP with SASL authentication and verification, enhancing email security for Unix and Linux environments.

Highlights:
- Compiled, installed and configured the open source openssl, perl, SASL, postfix
- Enable SMTP with SASL authentication for postfix
- Configured SASL authentication on the IMAP/POP3 Unix client side (SquirrelMail and pine)

Technologies:
Openssl-0.9.7, SASL-2.1, Postfix 2.4, SquirrelMail, pine, Solaris 9, Linux,

10.6 Installed AIX 5.1 and configured NIM on IBM RS6000 (10x 7043-140 boxes and 1x 7024-E30 box)

Installed and configured IBM AIX with NIM and NFS on RS6000 systems, enabling automated provisioning and dynamic resource management.

Highlights:

- Configured Network Interface and IP network
- Installed AIX 5.1 and configured NFS Server/Clients, NIS Clients
- Configured Network Installation Management (NIM)
- Deployed SORCER/Jini/Rio dynamic provisioning environment

Technologies:

AIX 5.1, NIM, NFS, NIS/NIS+, SORCER/Jini/Rio,

10.7 Deployed Globus Tookit3.2.1 (OGSA/OGSI) Grid Computing Environment on Solaris 9(3 boxes) and Linux (3 boxes)

Deployed and secured Globus Toolkit 3.2.1 grid computing environments on Solaris and Linux, supporting distributed research workloads with SSL certificates.

Highlights:

- Installed and configured OGSI core on Solaris 9 and Redhat Linux 9
- Installed OGSI GRAM, MDS, GSI, GridFTP and RLS on Solaris 9 and Red Hat Linux 9
- Configured distributing security (SSL certificate based) for Globus Toolkit 3.2.1
- Compiled, installed and Configured Perl on Solaris 9

Technologies:

Unix (Solaris 9, Redhat Linux 9), NFS, NIS/NIS+, Perl, PKI, X509, GNU Tools, Globus Toolkit 3.2.1, OGSA/OGSI, GSI,

10.8 Setup Jumpstart Environment for Solaris 9 and Solaris 10

Engineered automated Solaris deployments using Jumpstart, supporting multi-subnet installations and custom Perl scripts for NIS+ and tftp management.

Highlights:

- Configured tftp and bootparamd server for the Jumpstart client machine on Solaris 9 and Solaris 10 Server
- Diagnosed and debug the problems related to RARP/ARP
- Setup Jumpstart rules, profiles, sysidcfg configuration files, begin, install and post install scripts
- Setup NIS+ client configuration files in the Jumpstart
- Developed Perl scripts to maintain NIS+ tables and tftp boot parameters

Technologies:

Unix(Solaris 9), Jumpstart, TFTP, Bootparamd, NIS+, Perl, Security, snoop, RARP/ARP,

10.9 Designed and Implemented the Single Sign-On (SSO) for Unix and Windows Environments

Integrated Unix and Windows environments with a unified SSO solution using LDAP, Kerberos, AFS, and Samba, streamlining authentication and access control.

Highlights:

- Designed the Single-Sign-On (SSO) Solution for computer science department
- Researched the feasibility of different solutions of Single sign-on (SSO)
- Built the test environment for the solution in Solaris 8 and 9 platform
- Built the GNU compiling and make environment to make the source file of samba in Solaris 9 platform
- Tested the Microsoft Service for Unix 3.0 for our requirement
- Built and Configure Samba environment
- Developed a customized Kerberos and AFS client with AFS home drive mapping in windows environment
- Compiled OpenAFS, OpenLDAP and MIT Kerberos in UNIX (Linux and Solaris) environment
- Setup and Configure MIT Kerberos, OpenAFS and OpenLDAP in Unix (Linux, and Solaris) environment
- Rebuilt MIT Kerberos with krb524 in windows environment
- Developed PAM module for TMP dir environment variable setting
- Setup PAM modules configuration file (pam.conf) in UNIX (Debian Linux and Solaris) environments

Technologies:

Unix(Debian Linux, Solaris 8 and 9), LDAP(OpenLDAP2.1.2, SunONE Directory Server 5.2 , MS Active Directory), AFS (OpenAFS 1.2.10), Kerberos (SEAM 1.2, MIT Kerberos V 1.3.1). PAM, GSSAPI, SSL, SASL, SSH, C++, GNU make, GCC, Visual.NET, Perl,

10.10 The Migration from NIS+ to LDAP for Solaris

Migrated Solaris NIS+ directory services to LDAP, designing custom schemas and automation scripts to modernize user management and authentication.

Highlights:

- Installed and Configured the Solaris 8, Solaris 9 Environment
- Installed and configured the iPlanet Directory Server in Solaris platform
- Designed the LDAP schema for department's requirements
- Populated the NIS+ data to the LDAP
- Developed Perl scripts with LDAP enabled for user managements

Technologies:

Unix(Solaris 8 and 9), LDAP(SunONE Directory Server 5.1), Kerberos (SEAM 1.2). NIS/NIS+, SSL, SSH, Perl, GNU make , GCC,

11. Aug.,2002 – Jun.,2006, Master Student, Computer Science, Texas Tech University , Lubbock, Texas,USA

Studied as Master Student in the CS department while working as part time as System Administration for the Teaching and Research labs for Unix/ Linux and Windows environments

11.1 Master Thesis: A Service Wrapping and Provisioning Framework for SOA

Developed an automated SOA framework using Jini/Rio to wrap legacy applications as dynamic, provisioned services in a distributed grid environment, including a GIS-based yield tracker system.

Technologies:
Jini 2.0, RIO, JERI, Globus Toolkits 3.x, J2EE, Web Service, OpenGIS, PostGIS, JDBC, GeoServer, Struts, Ant, Tomcat, Servlet, JSP, UML2,

11.2 GIS Project: A Yield Mapping and Prediction Information Delivery System

Built a web-based GIS platform integrating OpenGIS, GeoServer, and PostGIS to deliver real-time yield prediction maps for farmers using satellite and field data.

Technologies:
OpenGIS, PostGIS, PostgreSQL, JDBC, GeoServer, GeoTools, Struts, Ant, Apache Maven, Servlet, JSP, MVC,

11.3 Independently finished practical course projects list

Completed advanced projects including a C-based LL(1) compiler, UML-modeled supermarket system, parallel matrix multiplication service, Linux kernel modules, Java card applications, and mobile map navigation tools.

Technologies:
GCC Compiler, Solaris, Linux Kernel 2.4, UML 2, Rational Rose, Java, SORCER, SOA, Jini, JavaSpace, JBoss, J2EE, VFS, Java Card platform, MIDP, MIDlets, Servlets, OCF, CLDC, Ant,

HOBBIES PROJECTS:

H.1 Personalized JSON Schema Resume

Customized the open source project dynamic-json-resume using JSON format and mustache templates with Node.js and CSS3. GitHub url: https://github.com/rongyj/dynamic-json-resume/tree/rongyj

Highlights:
- Adapt the JSON schema to personal resume format
- Developed new utility methods to filter the JSON data
- Personalized CSS3 styles with auto counting for multiple counters
- Add new table mustache template
- Add @media print page numbers to the CSS3 styles
- Add CSS3 style for large table crossing page breaks
- Debug in Atom with an open source Node-Debugger package
- Generate pdf file from html using puppeteer

Technologies:
JavaScript, HTML5, CSS3, Node.js, JSON, mustache, Puppeteer,